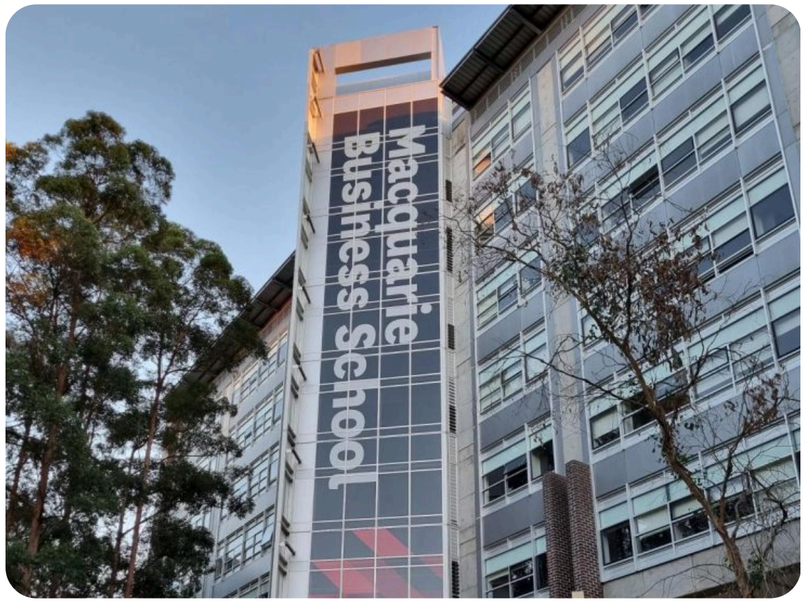
MACQUARIE University

Dashboard Solutions & Predictive Analytics

for MQBS Course Directors



Presented by
Group 7 Thursday 11:30 am



Aksara Chommueang
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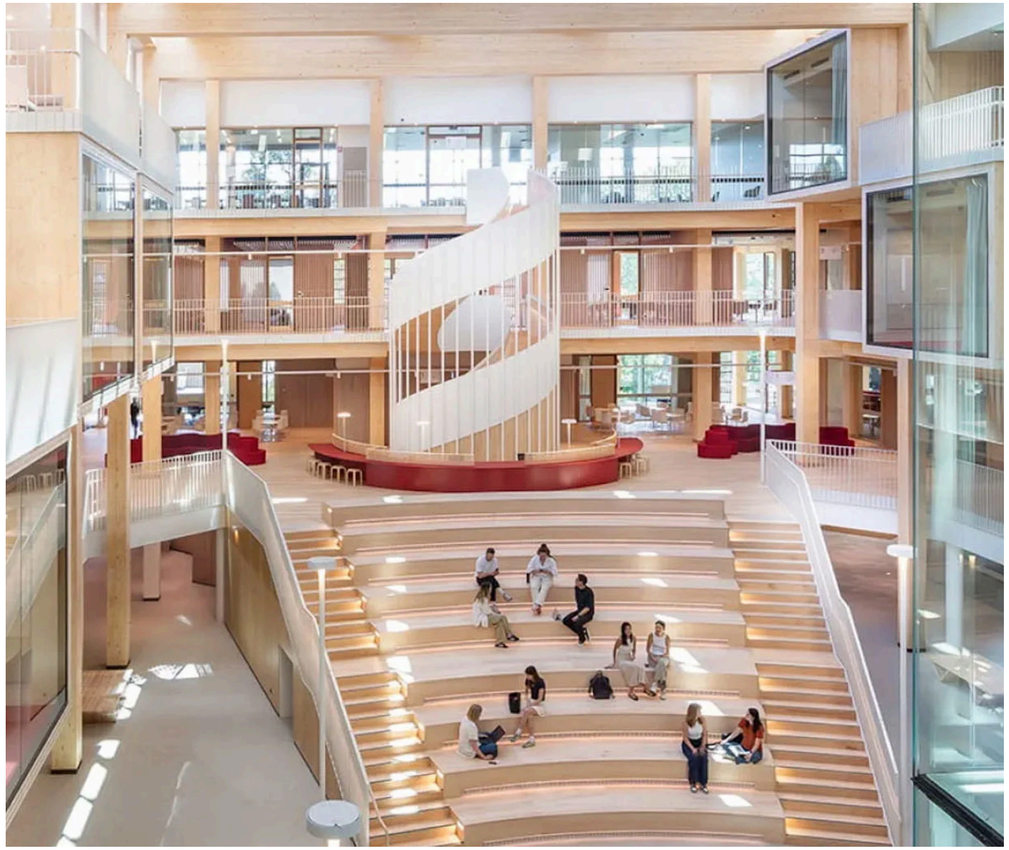


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Agenda



01. Why This Matters
02. Deliverables
03. Data Preparation
04. Dashboard Design
05. Dashboard 1
06. Dashboard 2
07. Predictive Modelling
08. Key Insights Summary
09. Recommendations

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Why This Matters



Improve student retention by shifting from reactive responses to **proactive, data-informed interventions**

For

Course Directors

Clear visibility into progression barriers and at-risk cohorts

For

The Students

Enhanced learning experiences and completion rates

For

The Institution

Stronger performance metrics and regulatory compliance

Deliverables



Dashboard 1:

Course Barrier

Dashboard 2:

Student Attrition & Risk Behaviour

Predictive Model:

Logistic Regression Model

Forecasting Dropout Risk

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Data Preparation

DATA SOURCE

De-identified enrolment and retention data of undergraduate students in Macquarie Business School (2020-2025)

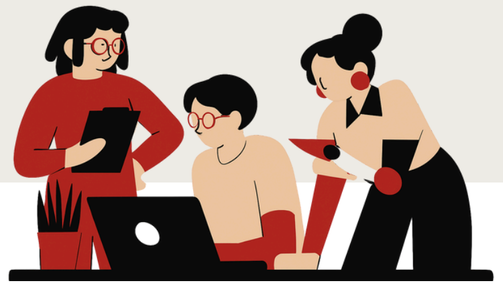


DATA PROCESSING

- Combined enrolment & retention datasets together
- Cleaned and standardised
- **Dashboards 1** used 2020-2025 records
- **Dashboard 2** used 2020-2024 records due to retention status data lag
- **Predictive Modelling**
  - **For training:**
    - Used 2020-2024 records with confirmed retention status
    - For continuing 2025 students, used 2024 retention records (2026 data unavailable)
    - Excluded students commencing in 2025
  - **For predicting:**
    - Used 2025 students data with “Not Retained in Year X+1” and new students

CORRELATION ANALYSIS

Conducted correlation analysis to identify the **most impactful features** for each research question and selected key variables for dashboard visualisation



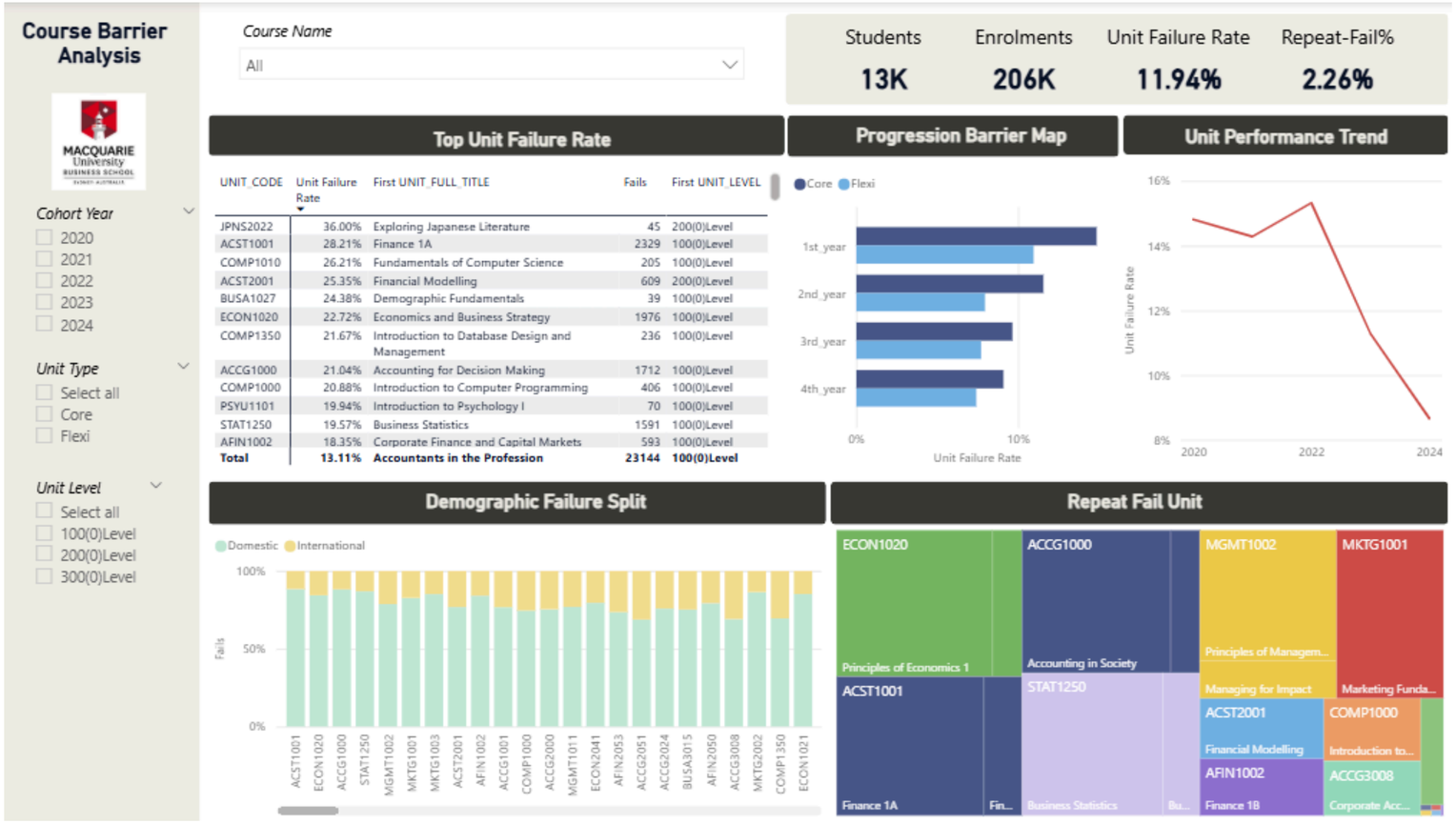
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Dashboard 1

Course Barrier

Persona: Course Director

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UNIT_CODE	Unit Failure Rate	First UNIT_FULL_TITLE	Fails	First UNIT_LEVEL
JPNS2022	36.00%	Exploring Japanese Literature	45	200(0)Level
ACST1001	28.21%	Finance 1A	2329	100(0)Level
COMP1010	26.21%	Fundamentals of Computer Science	205	100(0)Level
ACST2001	25.35%	Financial Modelling	609	200(0)Level
BUSA1027	24.38%	Demographic Fundamentals	39	100(0)Level
ECON1020	22.72%	Economics and Business Strategy	1976	100(0)Level
COMP1350	21.67%	Introduction to Database Design and Management	236	100(0)Level
ACCG1000	21.04%	Accounting for Decision Making	1712	100(0)Level
COMP1000	20.88%	Introduction to Computer Programming	406	100(0)Level
PSYU1101	19.94%	Introduction to Psychology I	70	100(0)Level
STAT1250	19.57%	Business Statistics	1591	100(0)Level
AFIN1002	18.35%	Corporate Finance and Capital Markets	593	100(0)Level
Total	**13.11%**	**Accountants in the Profession**	**23144**	**100(0)Level**

Purpose & Context

Address complex **structural challenges** within the curriculum

Identify where the **academic pressure peaks** within the curriculum's progression

Determine which units cause **persistent stagnation** and **repeat failures**

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Dashboard 1

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Course Barrier

Core Diagnostics: The What, Where, Why

Top Unit Failure Rate

UNIT_CODE	Unit Failure Rate	First UNIT_FULL_TITLE	Fails	First UNIT_LEVEL
COMP1010	30.00%	Fundamentals of Computer Science	147	100(0)Level
COMP1000	26.91%	Introduction to Computer Programming	169	100(0)Level
COMP1350	26.43%	Introduction to Database Design and Management	129	100(0)Level
AFIN1002	25.29%	Corporate Finance and Capital Markets	43	100(0)Level
ECON1031	24.29%	Quantitative Methods in Economics, Business and Finance	51	100(0)Level
STAT1250	20.24%	Business Statistics	102	100(0)Level
MGMT1002	18.92%	Managing for Impact	28	100(0)Level
ACST1001	18.55%	Finance 1A	59	100(0)Level
ECON1020	17.99%	Economics and Business Strategy	52	100(0)Level
ACCG1000	16.02%	Accounting for Decision Making	66	100(0)Level
COMP2350	14.94%	Database Systems	36	200(0)Level

Purpose

Provides prioritisation, finding the units causing the most damage by scale & severity

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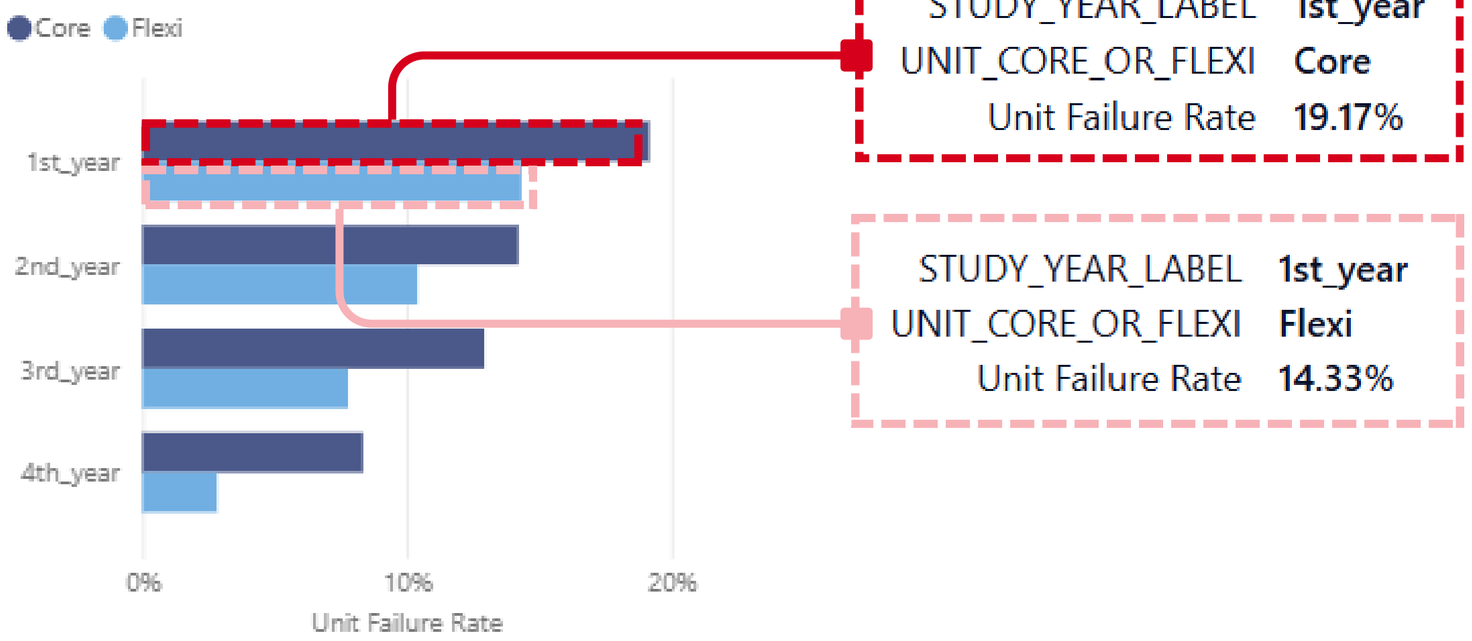
Dashboard 1

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Course Barrier

Core Diagnostics: The What, Where, Why

Progression Barrier Map



Purpose

Provides spatial context, locating where in the journey (e.g.1st Year) the pressure peaks

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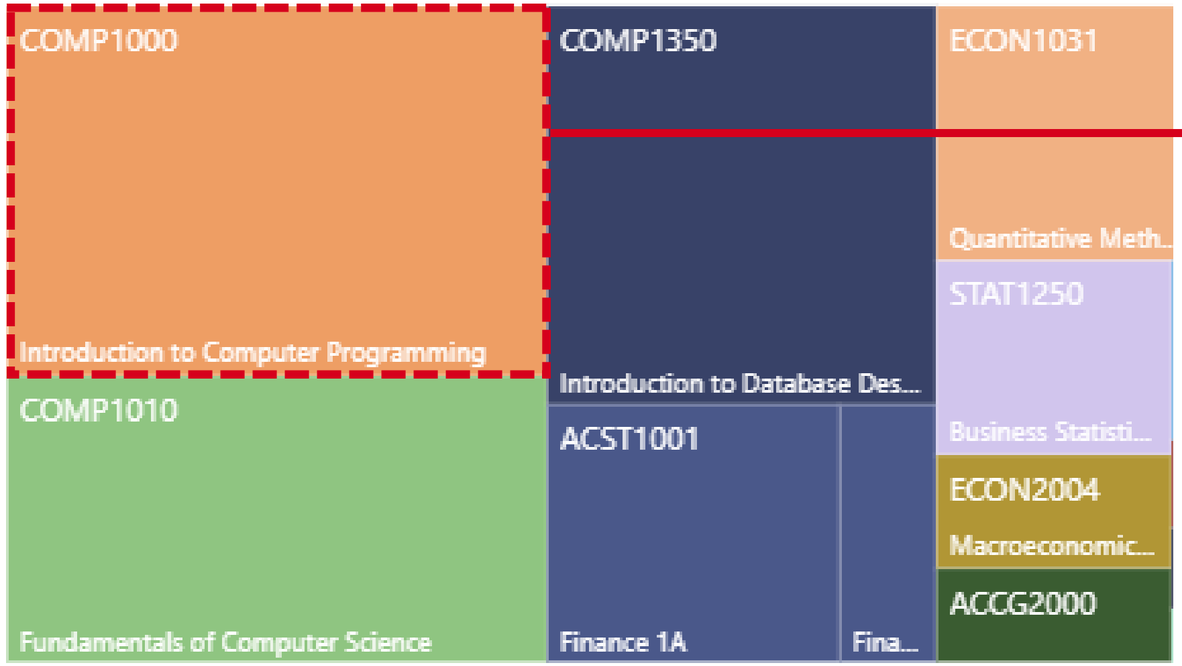
Dashboard 1

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Course Barrier

Core Diagnostics: The What, Where, Why

Repeat Fail Unit



UNIT_CODE	**COMP1000**
UNIT_FULL_TITLE	**Introduction to Computer Programming**
Count of REPEAT_FAIL	**628**
First UNIT_LEVEL	**100(0)Level**
Repeat-Fail %	**5.25%**

Purpose

Provides persistence check, proving the problem is structural, not a one-off failure

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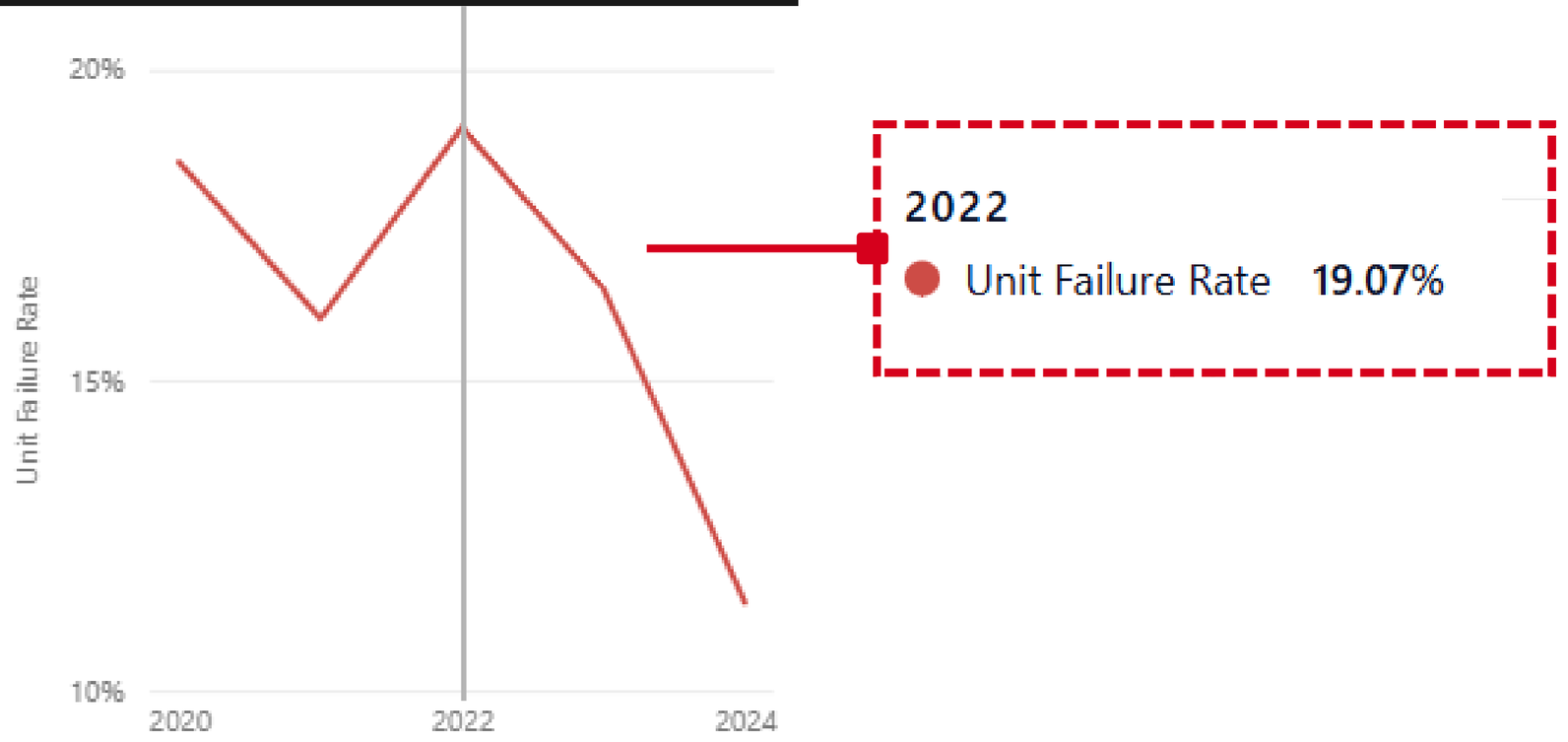
Dashboard 1

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Course Barrier

Core Diagnostics: The When & Who

Unit Performance Trend



Purpose

Provides stability of academic challenges over time. Separating persistent structural flaws or temporary anomalies

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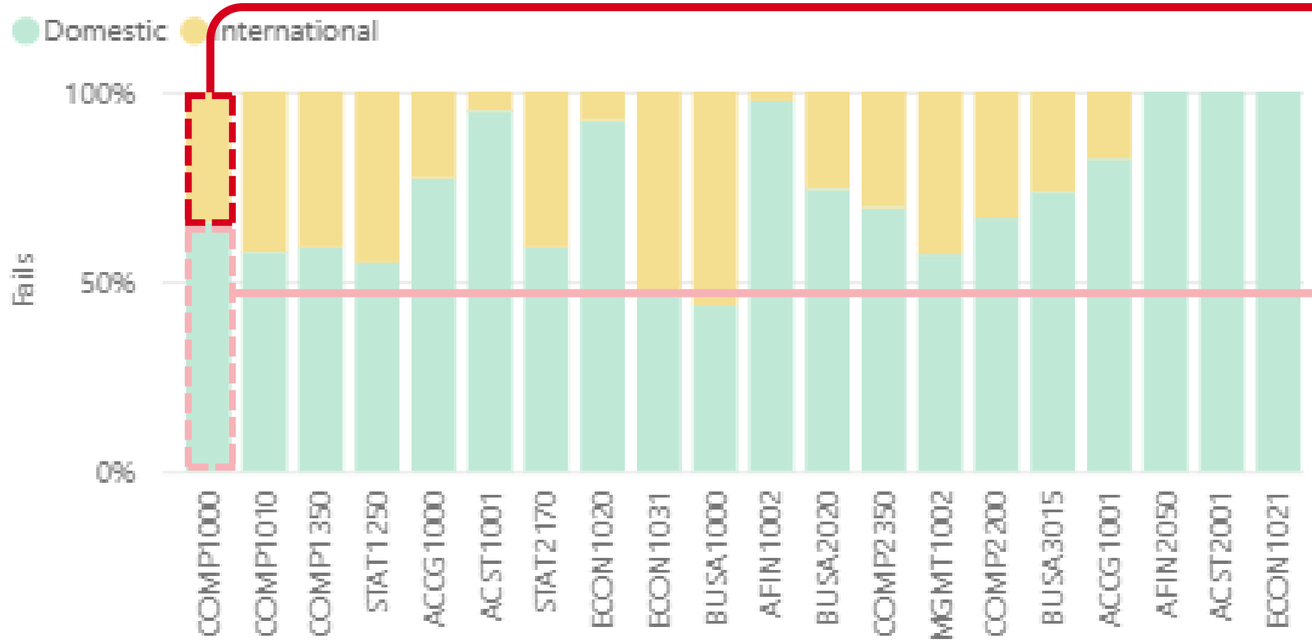
Dashboard 1

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Course Barrier

Core Diagnostics: The When & Who

Demographic Failure Split



UNIT_CODE	**COMP1000**
DOMESTIC_OR_INTERNATIONAL	**International**
Fails	**58 (34.32%)**
Fail %	**23.97%**
First UNIT_FULL_TITLE	**Introduction to Computer Programming**

UNIT_CODE	**COMP1000**
DOMESTIC_OR_INTERNATIONAL	**Domestic**
Fails	**111 (65.68%)**
Fail %	**28.76%**
First UNIT_FULL_TITLE	**Introduction to Computer Programming**

Purpose

Provides equity check, ensuring solutions are targeted and equitable, not one-size-fits-all

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Dashboard 1

Course Barrier

Value for Course Directors

1 Makes the Problem Manageable

Moves Course Directors from 50+ failing units > 2-3 high-priority structural barriers.

2 Ensures Targeted Interventions

Allows design for the specific level and specific demographic in greatest need.

3 Enables Proactive Curriculum Design

Provides the diagnostic tool to shift from reactive problem-solving to proactive, data-driven curriculum improvement.

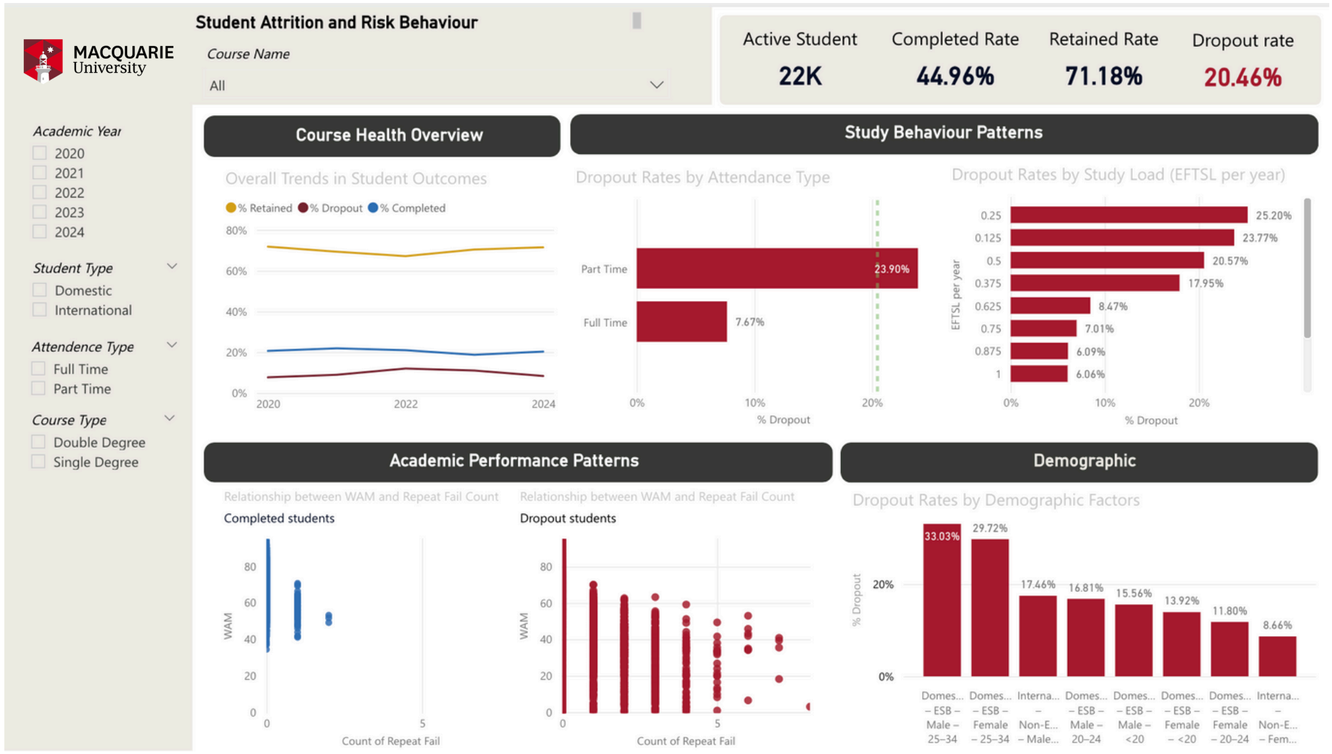
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Dashboard 2

Student Attrition & Risk Behaviour

Persona: Course Director

Group 7 Thursday 11:30am for MACQUARIE University



Purpose & Context

Designed to explore **behavioural risk**

Uncover **early warning signals,** revealing **patterns behind attrition**

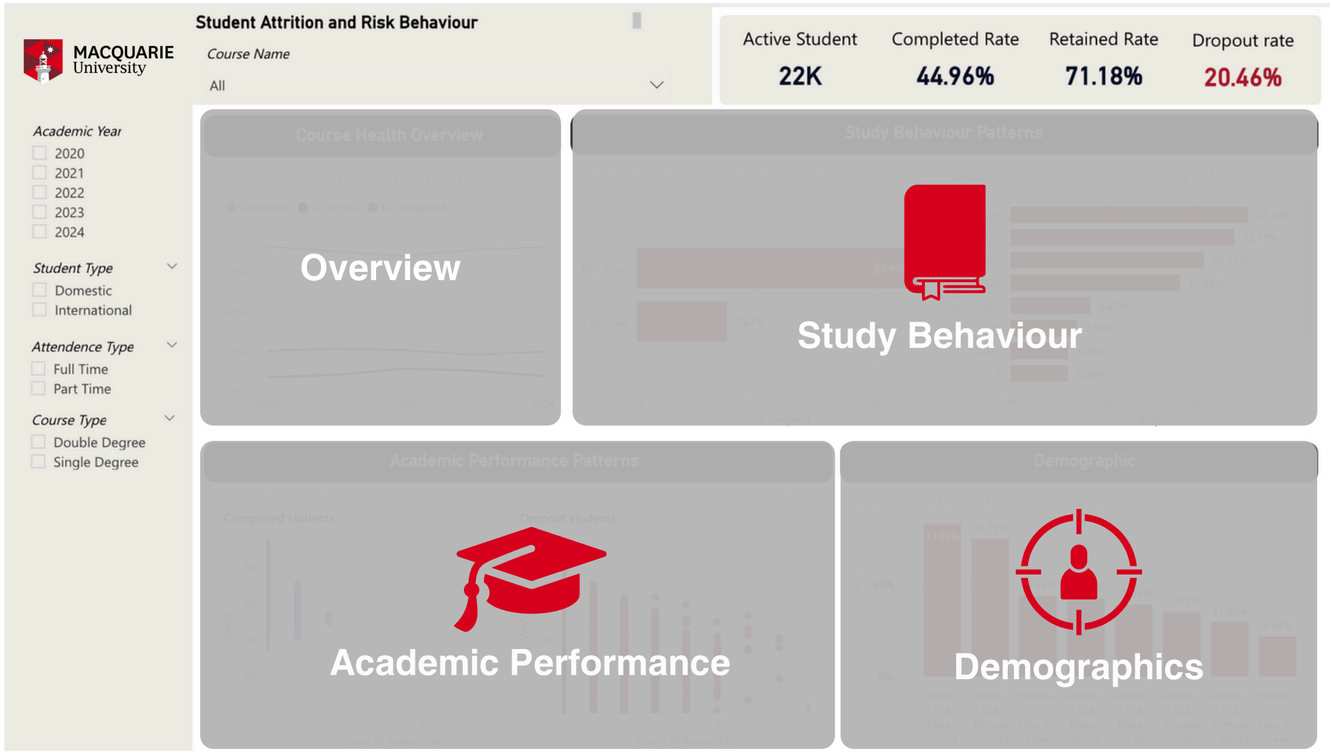
Integrate **study behaviour**, **academic performance** and **demographics**

Aksara Chommueang 47553618

Dashboard 2

Student Attrition & Risk Behaviour

Group 7 Thursday 11:30am for MACQUARIE University



Purpose & Context

- Designed to explore **behavioural risk**
- Uncover **early warning signals,** revealing **patterns behind attrition**
- Integrate **study behaviour**, **academic performance** and **demographics**

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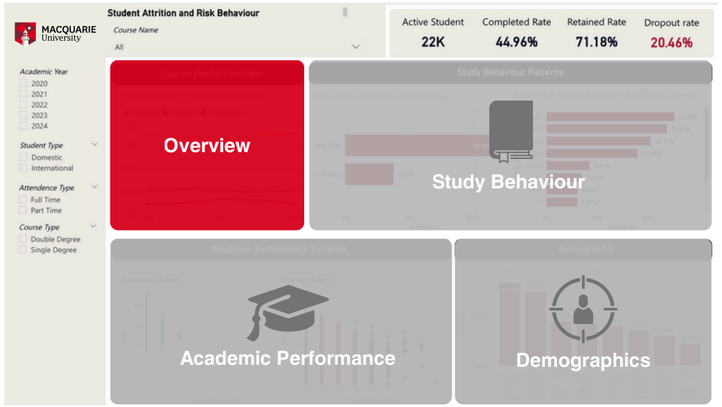
Dashboard 2

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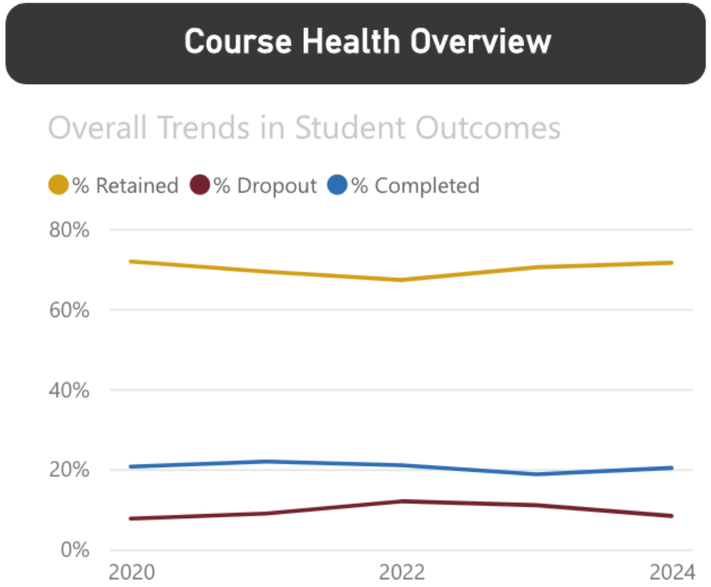
Student Attrition & Risk Behaviour

Overview

Interactive, real-time dashboard filtered by Course and Year providing course-specific insights.

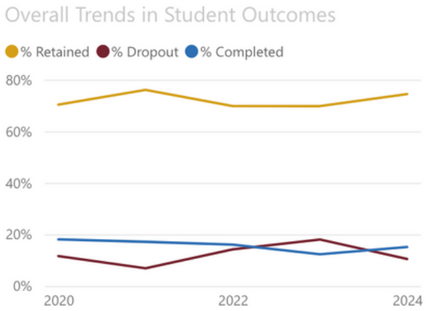


- **Starting point** for understanding course stability before drilling down into detailed risk factors

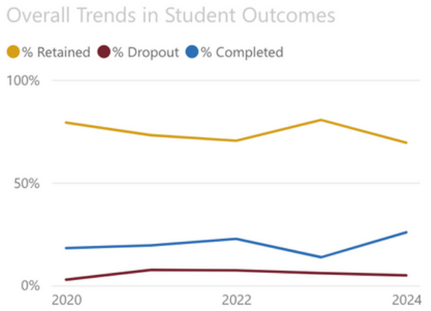


Overall Trends in Student Outcomes (2020-2024)

Example



Bachelor of Business Analytics



Bachelor of Marketing and Media

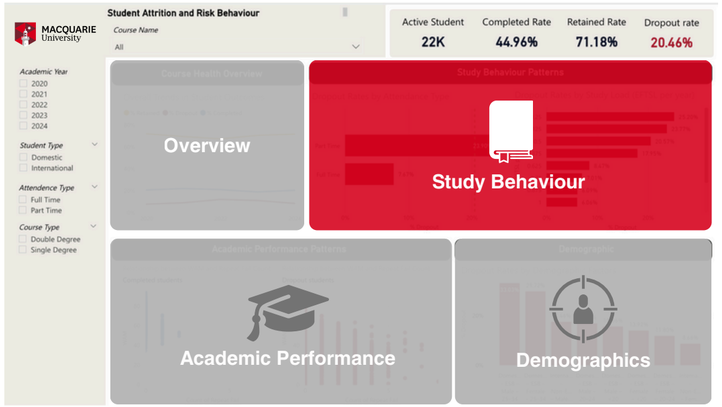
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Dashboard 2

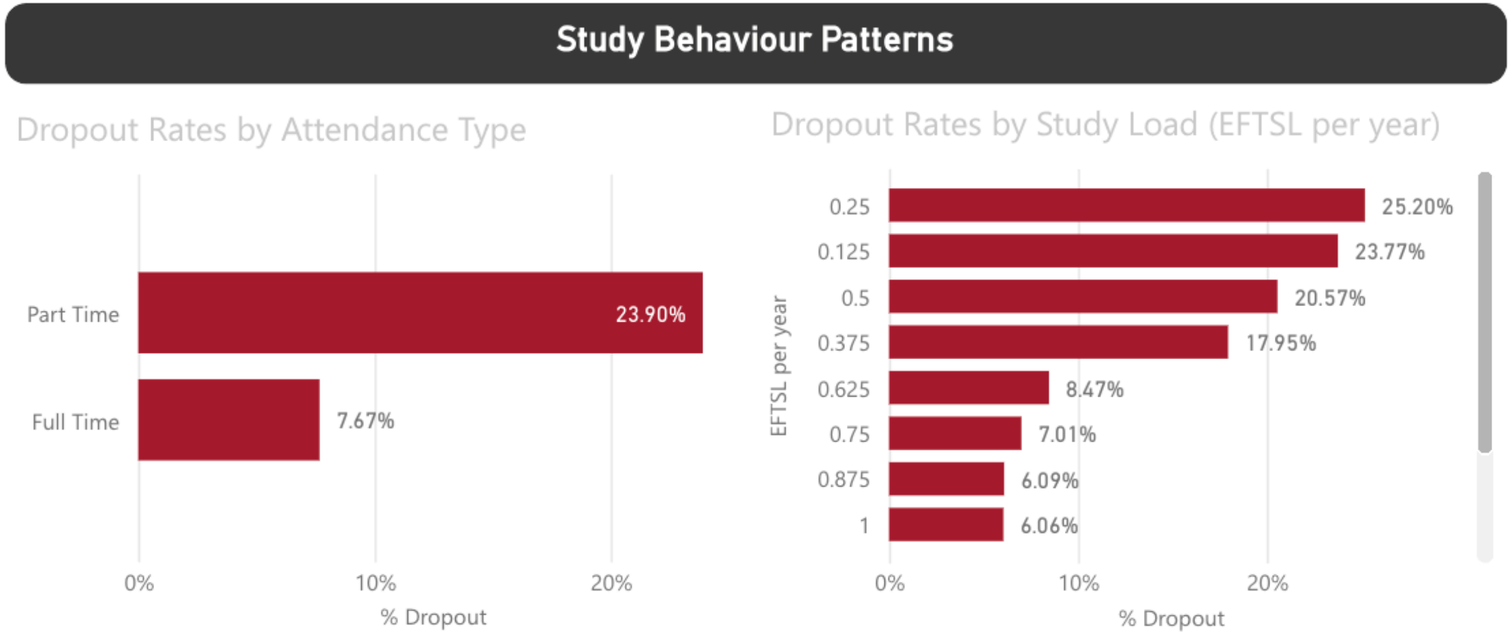
Student Attrition & Risk Behaviour

Study Behaviour

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- **Part-time students** and **students taking lighter loads** (<0.5) show the highest attrition
- This reveals that **reduced study intensity** is an early behavioural warning sign



Two bar charts comparing dropout rates by **attendance type** and **study load** (EFTSL per year)

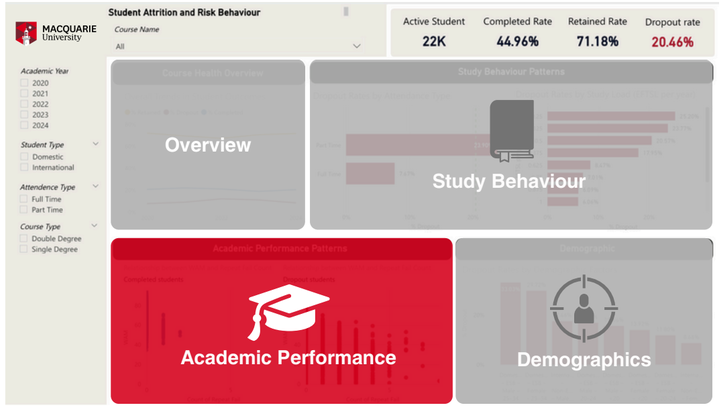
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Dashboard 2

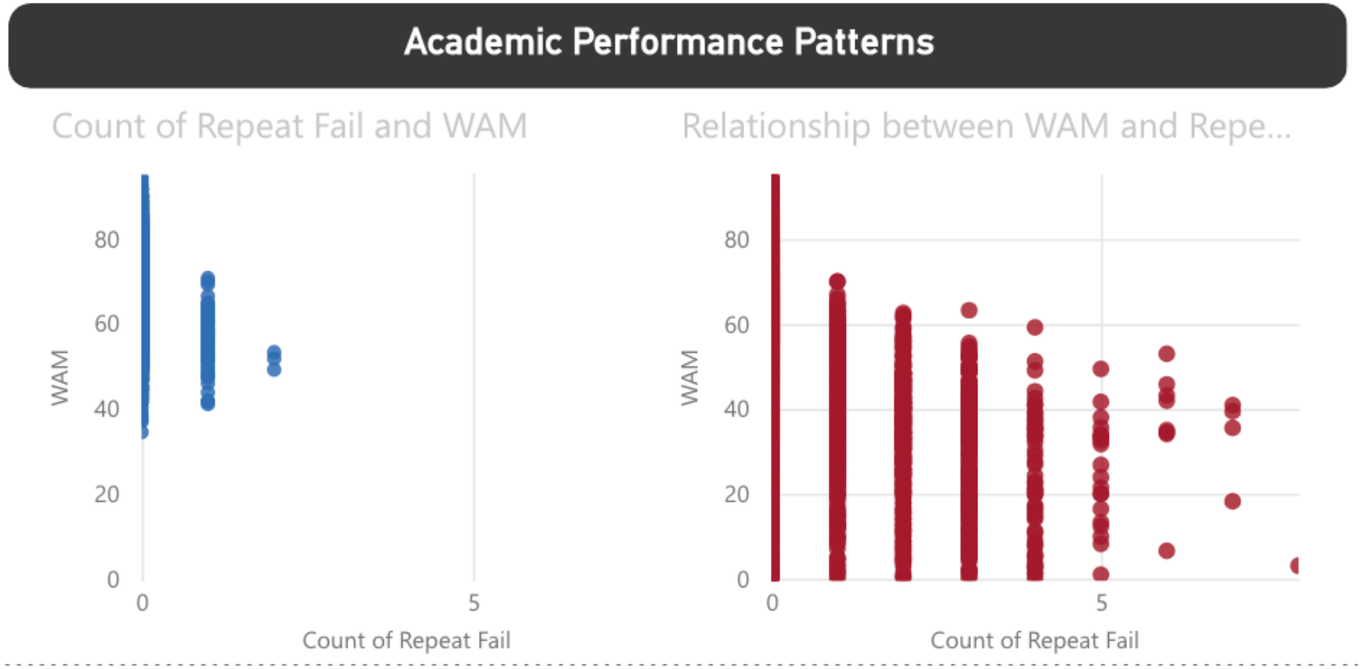
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Student Attrition & Risk Behaviour

Academic Performance



- Dropout students cluster at **low WAM with multiple repeated fails (>3)**
- It highlights that **attrition is not driven by academic ability alone** but motivation and engagement may also play a major role.



This scatter plot **relationship between WAM and repeat-fail counts** between **completed** and **dropout** groups

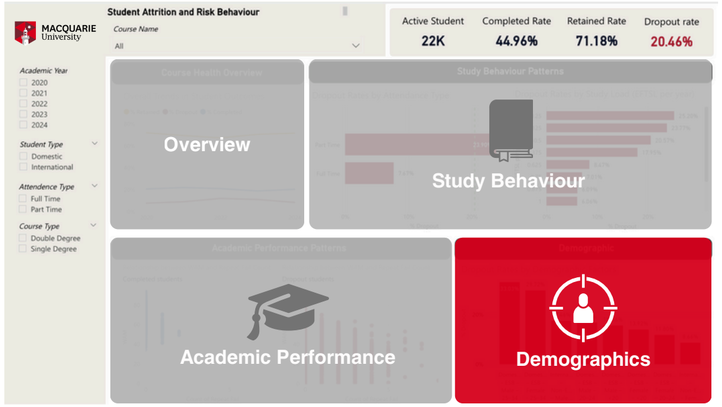
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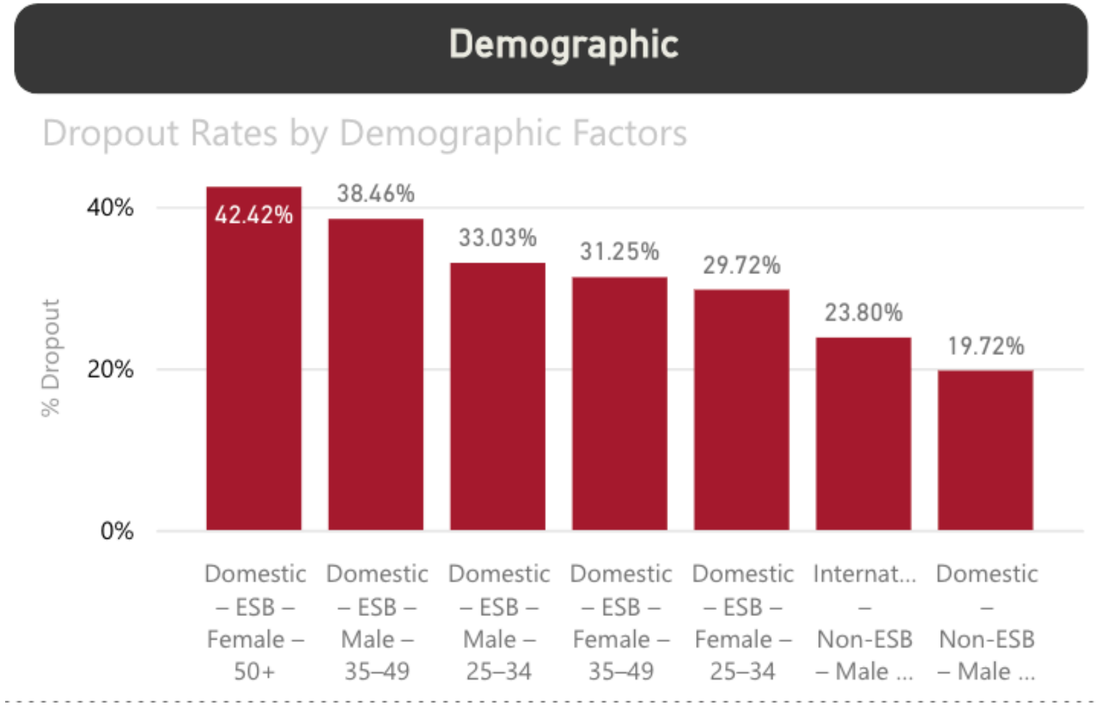
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Student Attrition & Risk Behaviour

Demographics



- **Domestic** and **English-speaking** cohorts <u>consistently</u> appear more likely to disengagement
- It points to demographic segments that may need to be explored further to understand the causes.



Demographic Differences Reveal At-Risk Student Profiles
(by nationality, language background, gender, and age group)

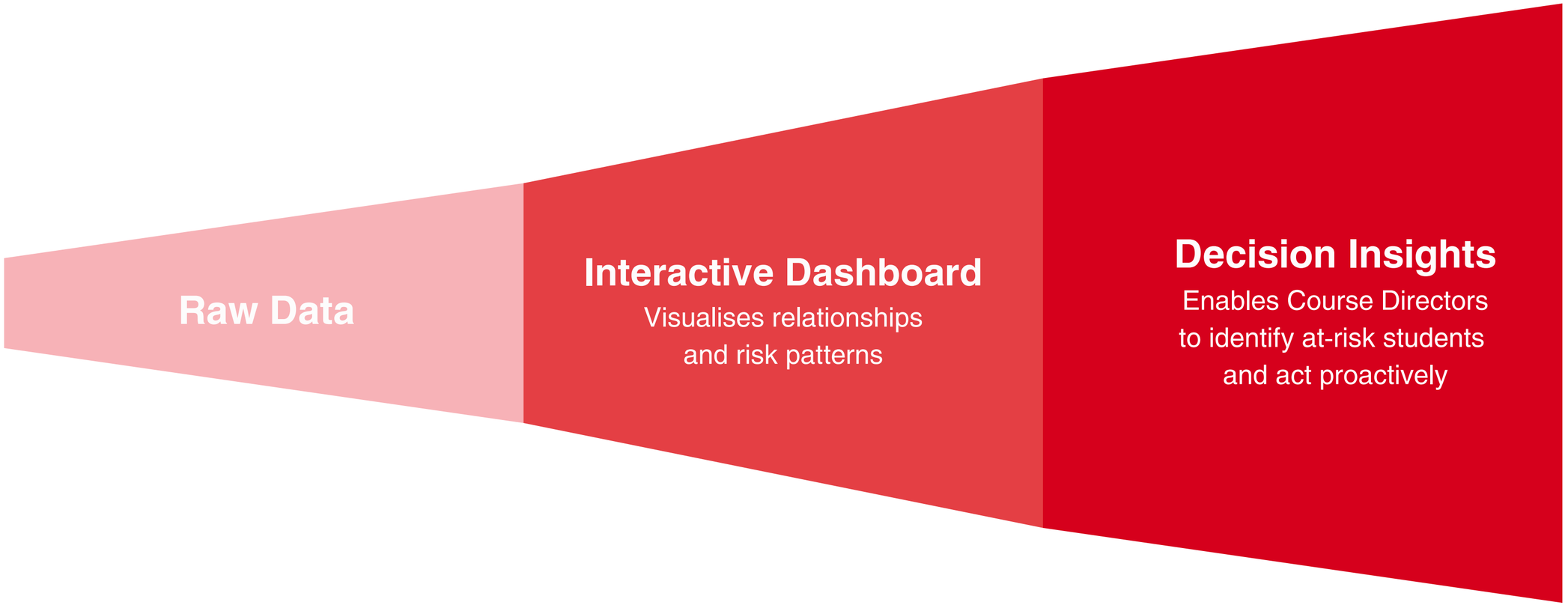
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Dashboard 2

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Student Attrition & Risk Behaviour

Value for Course Directors



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Predictive Modelling

PURPOSE



Predict which 2025 students are at risk of dropping out in 2026 to enable early intervention

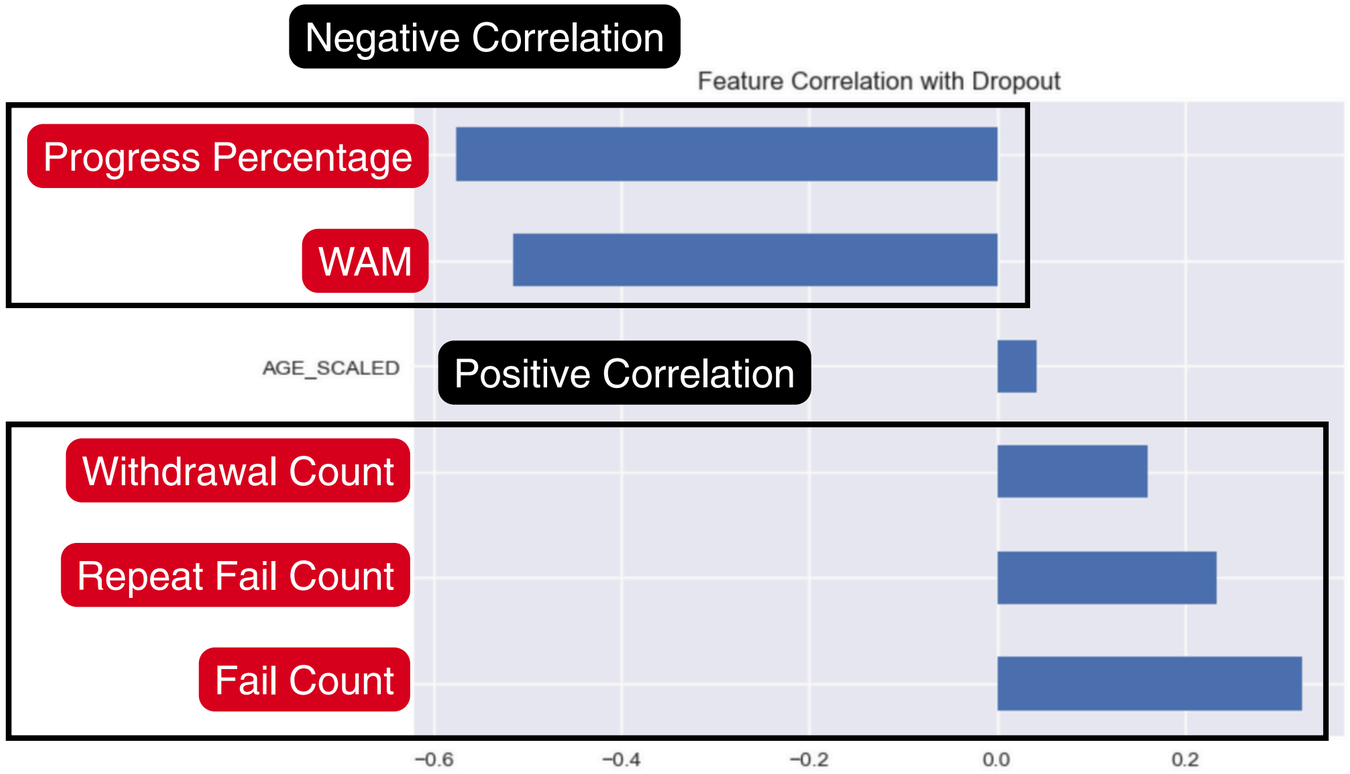
WHY LOGISTIC REGRESSION



- **Binary classification**
  – Retained vs Dropout
- **Interpretable coefficients**
  – Easy to communicate with non-technical stakeholders

Predictive Modelling

TOP 5 PREDICTIVE FEATURES



INSIGHTS

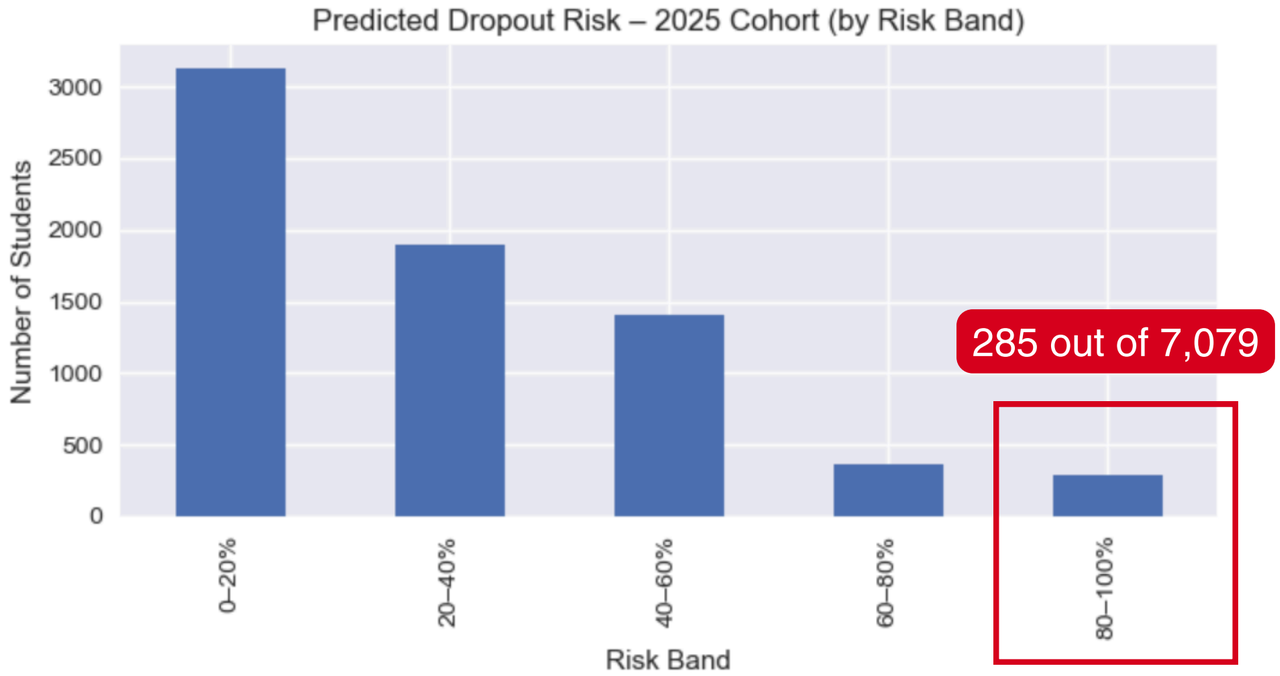
Students with **low progress** and **poor grades**, combined with **multiple failures**, face highest dropout risk

MODEL PERFORMANCE

86%	0.9
Accuracy	AUC

Supanida Poolsawat 47951923

Predictive Modelling



INSIGHTS

4% of active students in 2025 are likely to drop out in the next year

We set an alert threshold at 0.8 for predicted dropout probability.

Key Insights Summary

Foundation Units Need an Overhaul

- High failure rates indicate issues with 1st year course design or delivery.
- First-year students may struggle with the shift from high school to university.

4 Early Warning Signs Identified

- Key Risk Factors: Low WAM, Low Study Load, Low Progress, and Frequent Repeated Fails

Domestic ESB Needs Support

- Domestic students, particularly those from English-speaking backgrounds, show higher attrition risk.

Recommendations

1 Foundation Unit Review & Redesign

- Review high-failure course individually.
- Apply tailored redesign strategies: revise content, workload, or assessments/change UCs/reduce seminar sizes.

2 Enhance First-Year Transition Support

- Launch a “Student Mentors” program linking senior and first-year students.
- Include O-Week sessions on campus life and support services.

3 Implement Tiered Intervention

- Use predictive modeling to classify student risk levels
- Match support intensity to risk:

1. **High Risk:** mandatory academic coaching & wellbeing check-ins
2. **Medium Risk:** regular progress monitoring & peer group support
3. **Low Risk:** preventive care and light-touch guidance

Thank you!

